

AARUSH SHARMA

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Clarksville, Maryland



EDUCATION

University of Maryland, College Park
Bachelor of Science - Mechanical Engineering
Expected Graduation: May 2029

Relevant Coursework

Calculus III
Differential Equations
Physics : Mechanics and Particle Dynamics
Object-Oriented Programming I
Oral Communication : Principles and Practice

Activities - Taiwanese American Student Association, Coffee @ UMD,
Engineers Without Borders Organization

Atholton High School | Columbia, MD
Graduated - 2025

PROJECTS

RFID Implementation Project | 2021-Present

- Built a wireless tracking solution to improve inventory management efficiency.
- Deployed system in real warehouse environment to reduce search time and misplaced items.

Discord Music Bot | 2022 - 2023

- Built a voice-chat music bot using YouTube API; implemented command handling and queue logic.

Game Development Project | 2024 - Present

- Designed a 2D platformer game using Unity; implemented movement, collision, and scene logic.
 - *Currently on Break*

AWARDS

- Certification of Biliteracy
- UMD Honor Band Solo Bass Clarinetist
- Principal and Gold Honor Rolls
- 2nd Place MESA Mobile App (Regionals)
- 2nd Place MESA Community Cleanup (Regionals)

SKILLS

- Technical: Python, Java, Arduino, ESP32, Fusion 360, Excel, Git, RFID Systems, Wireless Communication Modules (ESP32, Arduino)
- Languages: English (Fluent), Hindi (Fluent)

PROFESSIONAL EXPERIENCE

Technical Intern

Wholesale Maryland KDS2 | 2021 - Present

- Managed inventory databases using Excel to track transactions, reducing misplaced products.
- Analyzed inventory trends to maintain accurate stock levels.
- Trained new hires on warehouse software and basic computer workflows.
- Supported English-language learners to improve workplace communication.

Engineering Intern

Johns Hopkins Applied Physics Laboratory (APL) | 2024 - 2025
Summer & Academic Year Internship

- Developed Python tools to collect and visualize data for hardware integration tests.
- Worked with RFID systems, embedded hardware, and wireless communication modules.
- Programmed Arduino and ESP32 microcontrollers for sensor and tracking prototypes.
- Designed 3D-printed enclosures and components using Fusion 360.

International Tutor

Education Matters For All (EMFA) | 2024 - 2025
International Non-profit Tutoring Organization

- Tutored students internationally online in math, English, science, and computer concepts.
- Created worksheets and educational videos to support remote learners.